14 $a = \sqrt{7} + \sqrt{c}$ and $b = \sqrt{63} + \sqrt{d}$ where c and d are positive integers.

Given that $c : d = 1 : 9$
find, in its simplest form, the ratio $a : b$

$d = 9c$, so $\sqrt{d} = 3\sqrt{c}$.

1 mark
Substitute d=9c

$b = \sqrt{63} + \sqrt{d} = 3\sqrt{7} + 3\sqrt{c}$.

$b = 3(\sqrt{7} + \sqrt{c}) = 3a$.

1 mark
Show b=3a

$a : b = 1 : 3$.

1 mark
1:3

(Total for Question 14 is 3 marks)